PADHAKU
Science · Class 10 Science



CHAPTER 2

Acids, Bases and Salts

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✨ Try each question yourself first, then check the answer. 💡
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⭐ Introduction

? **Someone in the family has acidity after overeating. Which would you suggest as a remedy – lemon juice, vinegar or baking soda solution? Which property did you think of?**

✔ ANSWER

Baking soda solution.

Acidity is caused by excess acid in the stomach. Baking soda is basic, so it neutralises (cancels) the excess acid. Lemon juice and vinegar are acids and would make it worse.

Do You Know? **What is litmus? Name some other natural indicators.**

✔ ANSWER

Litmus is a purple dye extracted from lichen (division Thallophyta). It is purple in a neutral solution.

Other natural indicators: red cabbage leaves, turmeric, coloured petals of Hydrangea, Petunia and Geranium.

❓ Questions · Introduction

Q 1 **You have three test tubes: distilled water, an acidic solution and a basic solution. If you are given only red litmus paper, how will you identify the contents of each test tube?**

✔ ANSWER

- Dip red litmus in each tube. The tube that turns red litmus **blue** has the **basic solution**.
- Now take this blue litmus paper (made blue by the base) and dip it in the other two tubes.
- The tube that turns it back to **red** has the **acidic solution**.
- The tube that causes no change in either test has **distilled water** (neutral).

⭐ Activities · 2.1.1 Acids and Bases in the Laboratory

Activity 2.1 **Test HCl, H_2SO_4, HNO_3, CH_3COOH, NaOH, $Ca(OH)_2$, KOH, $Mg(OH)_2$ and NH_4OH with red litmus, blue litmus, phenolphthalein and methyl orange. What change in colour did you observe? Tabulate (Table 2.1).**

✔ ANSWER

Sample	Red litmus	Blue litmus	Phenolphthalein	Methyl orange

HCl, H_2SO_4, HNO_3, CH_3COOH (acids)	No change	Turns red	Colourless (no change)	Turns red
NaOH, $Ca(OH)_2$, KOH, $Mg(OH)_2$, NH_4OH (bases)	Turns blue	No change	Turns pink	Turns yellow

Conclusion: Acids turn blue litmus red and methyl orange red; bases turn red litmus blue, phenolphthalein pink and methyl orange yellow.

Activity 2.2 **Test onion-treated cloth strips, vanilla essence and clove oil with dilute HCl and dilute NaOH. Which of these – vanilla, onion and clove – can be used as olfactory indicators?**

✔ ANSWER

Observation: The onion smell remains with HCl (acid) but disappears with NaOH (base). Vanilla smell remains with HCl but is not felt with NaOH. Clove oil smell remains with HCl but goes away with NaOH.

Conclusion: Onion, vanilla and clove can all be used as olfactory indicators – their odour is lost in a basic medium but stays in an acidic medium.

Activities · 2.1.2 How do Acids and Bases React with Metals?

Activity 2.3 **Add zinc granules to dilute sulphuric acid and pass the gas through soap solution.**

(i) What do you observe on the surface of zinc granules?

(ii) Why are bubbles formed in the soap solution?

(iii) What do you observe when a burning candle is brought near a gas-filled bubble?

(iv) Are the observations the same with HCl, HNO_3 and CH_3COOH?

✔ ANSWER

(i) Gas bubbles form on the surface of the zinc granules.

(ii) The gas evolved (hydrogen) gets trapped in the soap solution, forming bubbles.

(iii) The gas burns with a **pop sound** – this shows the gas is **hydrogen**.

(iv) With HCl and CH_3COOH the observations are the same (hydrogen is given off; with acetic acid the reaction is slower). With HNO_3, hydrogen is usually not obtained, because nitric acid is a strong oxidising agent and oxidises the H_2 formed to water.

$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$

? **Acid + Metal → Salt + Hydrogen gas. Can you now write the equations for the reactions you have observed?**

✔ ANSWER

$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$

$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

$Zn(s) + 2CH_3COOH(aq) \rightarrow (CH_3COO)_2Zn(aq) + H_2(g)$

Activity 2.4 **Warm zinc granules with sodium hydroxide solution and repeat the steps of Activity 2.3. Record your observations.**

✔ ANSWER

Observation: Bubbles of gas are given off; the gas burns with a pop sound near a candle, so it is hydrogen.

$2NaOH(aq) + Zn(s) \rightarrow Na_2ZnO_2(s) + H_2(g)$ (sodium zincate)

Conclusion: Some metals like zinc react with bases also to give hydrogen gas and a salt.

⭐ Activities · 2.1.3 How do Metal Carbonates and Metal Hydrogencarbonates React with Acids?

Activity 2.5 **Add dilute HCl to sodium carbonate (tube A) and sodium hydrogencarbonate (tube B). What do**

you observe? Pass the gas through lime water and record your observations.

✔ ANSWER

Observation: Brisk effervescence (bubbling) occurs in both tubes. The gas turns lime water **milky**; with excess gas the milkiness disappears.

A: $Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$

B: $NaHCO_3(s) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l) + CO_2(g)$

$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$ (white precipitate)

Conclusion: Metal carbonates and hydrogencarbonates react with acids to give a salt, water and carbon dioxide.

⭐ Activities · 2.1.4 How do Acids and Bases React with each other?

Activity 2.6 **Add phenolphthalein to dilute NaOH, then add dilute HCl drop by drop, then a few drops of NaOH again.**

(i) What is the colour of the solution?

(ii) Is there any colour change on adding HCl?

(iii) Why did the colour of phenolphthalein change after adding acid?

(iv) Does the pink colour reappear on adding NaOH? Why?

✔ ANSWER

(i) **Pink** (phenolphthalein is pink in a base).

(ii) Yes, the pink colour slowly fades and the solution becomes **colourless**.

(iii) The acid neutralises the base, so no base is left; phenolphthalein is colourless in neutral and acidic solutions.

(iv) **Yes.** The added NaOH makes the solution basic again, so phenolphthalein turns pink again.

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$ (neutralisation)

Activities · 2.1.5 Reaction of Metallic Oxides with Acids

Activity 2.7 **Add dilute HCl slowly to copper oxide while stirring. Note the colour of the solution. What has happened to the copper oxide? Write and balance the equation.**

✔ ANSWER

Observation: The black copper oxide dissolves and the solution becomes **blue-green** due to copper(II) chloride.

$CuO(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + H_2O(l)$

Conclusion: Metal oxide + Acid → Salt + Water. So metallic oxides are **basic oxides**.

Questions · 2.1 Understanding the Chemical Properties of Acids and Bases

Q 1 **Why should curd and sour substances not be kept in brass and copper vessels?**

✔ ANSWER

Curd and sour substances contain **acids** (e.g. lactic acid). These acids react with copper and brass (an alloy of copper and zinc) to form poisonous compounds. This spoils the food and can cause food poisoning. So they are kept in glass or steel vessels.

Q 2 **Which gas is usually liberated when an acid reacts with a metal? Illustrate with an example. How will you test for the presence of this gas?**

✔ ANSWER

Hydrogen gas is liberated.

Example: $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$

Test: Bring a burning candle (or matchstick) near the gas. It burns with a **pop sound**, which shows it is hydrogen.

Q 3 **Metal compound A reacts with dilute hydrochloric acid to produce effervescence. The gas evolved extinguishes a burning candle. Write a balanced chemical equation for the reaction if one of the compounds formed is calcium chloride.**

✔ ANSWER

The gas that puts out a burning candle is CO_2, and the salt is calcium chloride, so **A is calcium carbonate ($CaCO_3$)**.

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

⭐ Activities · 2.2 What do all Acids and all Bases have in Common?

Activity 2.8 **Pass current through dilute HCl, dilute H_2SO_4, glucose and alcohol solutions using a bulb circuit.**

What do you observe? Does the bulb glow in all cases?

Repeat with alkalis like NaOH and $Ca(OH)_2$. What can you conclude?

✔ ANSWER

Observation: The bulb glows with dilute HCl and H_2SO_4, but **does not glow** with glucose and alcohol solutions.

Conclusion: Acids produce ions (H^+) in water, which carry current. Glucose and alcohol contain hydrogen but do not give H^+ ions, so they are not acids.

With alkalis (NaOH, $Ca(OH)_2$) the bulb also glows: bases also give ions (OH^-) in water and conduct electricity.

Activity 2.9 **Add concentrated H_2SO_4 to solid NaCl and test the gas with dry and wet blue litmus paper.**

Is there a gas coming out? In which case does the litmus change colour?

What do you infer about the acidic character of (i) dry HCl gas (ii) HCl solution?

✔ ANSWER

Observation: HCl gas comes out of the delivery tube. **Dry** blue litmus does not change colour; **wet** blue litmus turns **red**.

(i) **Dry HCl gas** is **not acidic** – it does not give H^+ ions without water.

(ii) **HCl solution** is **acidic** – in water, HCl gives H^+ (H_3O^+) ions: $HCl + H_2O \rightarrow H_3O^+ + Cl^-$

Do You Know? **Which of the bases in Table 2.1 are alkalis?**

✔ ANSWER

Alkalis are bases that dissolve in water: **NaOH, KOH, $Ca(OH)_2$ and NH_4OH**. ($Ca(OH)_2$ is only slightly soluble.) $Mg(OH)_2$ is almost insoluble in water, so it is generally not called an alkali.

Activity 2.10 **Add a few drops of concentrated H_2SO_4 to 10 mL water and touch the base of the beaker. Repeat with sodium hydroxide pellets.**
Is there a change in temperature? Is this an exothermic or endothermic process?

✔ ANSWER

Observation: The beaker becomes **hot** in both cases.

Conclusion: Dissolving an acid or a base in water is a highly **exothermic** process. So acid must always be added slowly to water with constant stirring.

? Questions · 2.2 What do all Acids and all Bases have in Common?

Q 1 **Why do HCl, HNO_3, etc., show acidic characters in aqueous solutions while solutions of compounds like alcohol and glucose do not show acidic character?**

✔ ANSWER

HCl, HNO_3 etc. **give H^+(aq) ions** in water, and H^+ ions are responsible for acidic properties.

Alcohol and glucose contain hydrogen, but they **do not ionise** in water to give H^+ ions. So they do not show acidic character.

Q 2 **Why does an aqueous solution of an acid conduct electricity?**

✔ ANSWER

An acid in water produces **ions** – $H^+(aq)$ (H_3O^+) and anions such as Cl^-. These moving ions carry electric current, so the solution conducts electricity.

Q 3 **Why does dry HCl gas not change the colour of the dry litmus paper?**

✔ ANSWER

Acidic behaviour is due to H^+ ions. Dry HCl gas does **not** give H^+ ions because separation of H^+ from HCl happens only in the presence of **water**. With no H^+ ions, dry litmus does not change colour.

Q 4 **While diluting an acid, why is it recommended that the acid should be added to water and not water to the acid?**

✔ ANSWER

Mixing a concentrated acid with water is **highly exothermic**.

If water is added to the concentrated acid, the large heat produced may make the mixture **splash out and cause burns**, and the glass container may break due to local heating.

Adding acid slowly to a large amount of water with stirring spreads the heat safely.

Q 5 **How is the concentration of hydronium ions (H_3O^+) affected when a solution of an acid is diluted?**

✔ ANSWER

On dilution, the concentration of H_3O^+ ions per unit volume **decreases**.

Q 6 **How is the concentration of hydroxide ions (OH^-) affected when excess base is dissolved in a solution of sodium hydroxide?**

✔ ANSWER

The concentration of OH^- ions per unit volume **increases** (up to the point the solution is saturated).

⭐ Activities · 2.3 How Strong are Acid or Base Solutions?

Activity 2.11 **Test the pH of the solutions in Table 2.2 and record the nature of each substance.**

💡 SAMPLE ANSWER

Approximate values (colours of pH paper may vary):

S. No.	Solution	Approx. pH	Nature
1	Saliva (before meal)	about 7.4	Slightly basic (nearly neutral)
2	Saliva (after meal)	about 5.8	Acidic
3	Lemon juice	about 2.2	Acidic
4	Colourless aerated drink	about 3–4	Acidic
5	Carrot juice	about 6	Weakly acidic
6	Coffee	about 5	Acidic
7	Tomato juice	about 4	Acidic
8	Tap water	about 7	Neutral
9	1M NaOH	14	Strongly basic
10	1M HCl	0	Strongly acidic

? **Acids that give more H^+ ions are strong acids and those that give less H^+ ions are weak acids. Can you now say what weak and strong bases are?**

✔ ANSWER

Bases that give **more OH^- ions** in water are **strong bases** (e.g. NaOH, KOH).

Bases that give **less OH^- ions** are **weak bases** (e.g. NH_4OH).

Do You Know? **The atmosphere of Venus is made up of thick clouds of sulphuric acid. Do you think life can exist on this planet?**

💡 SAMPLE ANSWER

No. Living organisms can survive only in a narrow range of pH (our body works at pH 7.0–7.8). Clouds of sulphuric acid make the conditions far too acidic for life as we know it.

Activity 2.12 **Shake 2 g soil with 5 mL water, filter and check the pH of the filtrate with universal indicator paper. What can you conclude about the ideal soil pH for the growth of plants in your region?**

💡 SAMPLE ANSWER

Observation: The universal indicator shows the pH of the soil, e.g. about 6.5–7.5 for most farm soil.

Conclusion: Most plants grow best in soil that is nearly neutral (pH about 6–7.5). If the soil is too acidic or too basic, it must be treated before growing crops.

Do You Know? **Nettle sting is due to methanoic acid; rubbing with a dock leaf gives relief. Can you guess the nature of the dock plant? Are you aware of any other traditional remedies for such stings?**

💡 SAMPLE ANSWER

The dock plant is **basic** in nature – it neutralises the methanoic acid of the sting.

Other remedies: applying baking soda paste on a bee or ant sting; applying a mild base like lime water or soap.

? Questions · 2.3 How Strong are Acid or Base Solutions?

Q 1 **You have two solutions, A and B. The pH of solution A is 6 and pH of solution B is 8. Which solution has more hydrogen ion concentration? Which of this is acidic and which one is basic?**

✔ ANSWER

Lower pH means higher H^+ concentration, so **solution A (pH 6)** has more hydrogen ion concentration.

A is acidic (pH < 7) and **B is basic** (pH > 7).

Q 2 **What effect does the concentration of $H^+(aq)$ ions have on the nature of the solution?**

✔ ANSWER

The **higher** the $H^+(aq)$ ion concentration, the **more acidic** the solution (lower pH).

As H^+ concentration decreases, the solution becomes less acidic, then neutral, then basic.

Q 3 **Do basic solutions also have $H^+(aq)$ ions? If yes, then why are these basic?**

✔ ANSWER

Yes, basic solutions also have $H^+(aq)$ ions (from water).

They are basic because the concentration of **OH^- ions is more than** that of H^+ ions.

Q 4 **Under what soil condition do you think a farmer would treat the soil of his fields with quick lime (calcium oxide) or slaked lime (calcium hydroxide) or chalk (calcium carbonate)?**

✔ ANSWER

When the soil is **too acidic** (pH too low). Quick lime, slaked lime and chalk are basic, so they neutralise the excess acid and bring the soil pH to the range suitable for crops.

⭐ Activities · 2.4 More about Salts

Activity 2.13 (a) **Write the chemical formulae of: potassium sulphate, sodium**

sulphate, calcium sulphate, magnesium sulphate, copper sulphate, sodium chloride, sodium nitrate, sodium carbonate and ammonium chloride. Identify the acids and bases from which they may be obtained.

✔ ANSWER

Salt	Formula	Acid	Base
Potassium sulphate	K_2SO_4	H_2SO_4	KOH
Sodium sulphate	Na_2SO_4	H_2SO_4	NaOH
Calcium sulphate	$CaSO_4$	H_2SO_4	$Ca(OH)_2$
Magnesium sulphate	$MgSO_4$	H_2SO_4	$Mg(OH)_2$
Copper sulphate	$CuSO_4$	H_2SO_4	$Cu(OH)_2$
Sodium chloride	NaCl	HCl	NaOH
Sodium nitrate	$NaNO_3$	HNO_3	NaOH
Sodium carbonate	Na_2CO_3	H_2CO_3	NaOH
Ammonium chloride	NH_4Cl	HCl	NH_4OH

Activity 2.13 (b) **Salts with the same positive or negative radical belong to a family. How many families can you identify among the salts given in this Activity?**

✔ ANSWER

- **Sulphate family:** K_2SO_4, Na_2SO_4, $CaSO_4$, $MgSO_4$, $CuSO_4$

- **Sodium family:** Na_2SO_4, NaCl, $NaNO_3$, Na_2CO_3
- **Chloride family:** NaCl, NH_4Cl

So there are mainly **3 families** (sulphate, sodium and chloride) having more than one member. (Each other ion – potassium, calcium, magnesium, copper, nitrate, carbonate, ammonium – appears in only one salt here.)

Activity 2.14 **Check the solubility and pH of sodium chloride, potassium nitrate, aluminium chloride, zinc sulphate, copper sulphate, sodium acetate, sodium carbonate and sodium hydrogencarbonate. Which are acidic, basic or neutral? Identify the acid and base used. Report in Table 2.4.**

✔ ANSWER

All these salts dissolve in water.

Salt	pH	Acid used	Base used
Sodium chloride	7 (neutral)	HCl	NaOH
Potassium nitrate	7 (neutral)	HNO_3	KOH
Aluminium chloride	< 7 (acidic)	HCl	$Al(OH)_3$
Zinc sulphate	< 7 (acidic)	H_2SO_4	$Zn(OH)_2$

Copper sulphate	< 7 (acidic)	H_2SO_4	$Cu(OH)_2$
Sodium acetate	> 7 (basic)	CH_3COOH	NaOH
Sodium carbonate	> 7 (basic)	H_2CO_3	NaOH
Sodium hydrogencarbonate	> 7 (basic)	H_2CO_3	NaOH

Conclusion: Strong acid + strong base → neutral salt (pH 7); strong acid + weak base → acidic salt (pH < 7); strong base + weak acid → basic salt (pH > 7).

? **Did you check the pH of sodium hydrogencarbonate in Activity 2.14? Can you correlate why it can be used to neutralise an acid?**

✔ ANSWER

Its pH is **more than 7**, so it is a mild, non-corrosive **basic** salt. Being basic, it reacts with and neutralises acids.

? **What does $10H_2O$ signify in $Na_2CO_3.10H_2O$? Does it make Na_2CO_3 wet?**

✔ ANSWER

$10H_2O$ means **10 molecules of water of crystallisation** are present in one formula unit of washing soda.

No, it does not make it wet. This water is part of the crystal structure, so the crystals are dry to touch.

Activity 2.15 **Heat a few crystals of copper sulphate in a dry boiling tube, then add 2–3 drops of water.**

What is the colour after heating? Do you notice water droplets? Where have these come from? Is the blue colour restored?

✔ ANSWER

After heating, the blue crystals turn **white** (anhydrous $CuSO_4$).

Water droplets appear on the cooler walls of the tube. They come from the **water of crystallisation** of the crystals.

On adding water, the **blue colour is restored**.

$CuSO_4.5H_2O$ (blue) → (heat) $CuSO_4$ (white) + $5H_2O$

? **Try to find out why calcium sulphate hemihydrate is called 'Plaster of Paris'.**

💡 SAMPLE ANSWER

It is called Plaster of Paris because it was first made on a large scale from gypsum found in large deposits near **Paris** (France).

❓ Questions · 2.4 More about Salts

Q 1 **What is the common name of the compound $Ca(ClO)_2$?**

✔ ANSWER

Bleaching powder.

Q 2 **Name the substance which on treatment with chlorine yields bleaching powder.**

✔ ANSWER

Dry slaked lime, $Ca(OH)_2$ (calcium hydroxide).

$2Ca(OH)_2 + 2Cl_2 \rightarrow Ca(ClO)_2 + CaCl_2 + 2H_2O$

Q 3 **Name the sodium compound which is used for softening hard water.**

✔ ANSWER

Washing soda – sodium carbonate ($Na_2CO_3.10H_2O$).

Q 4 **What will happen if a solution of sodium hydrocarbonate is heated? Give the equation of the reaction involved.**

✔ ANSWER

Sodium hydrogencarbonate breaks down to form **sodium carbonate, water and carbon dioxide** gas.

$2NaHCO_3 \rightarrow$ (heat) $Na_2CO_3 + H_2O + CO_2$

Q 5 **Write an equation to show the reaction between Plaster of Paris and water.**

✔ ANSWER

$CaSO_4.½H_2O + 1½H_2O \rightarrow CaSO_4.2H_2O$

(Plaster of Paris) → (Gypsum, a hard solid mass)

📝 Exercises

1 **A solution turns red litmus blue, its pH is likely to be (a) 1 (b) 4 (c) 5 (d) 10**

✔ ANSWER

(d) 10 – it turns red litmus blue, so it is basic, and basic solutions have pH more than 7.

2 **A solution reacts with crushed egg-shells to give a gas that turns lime-water milky. The solution contains (a) NaCl (b) HCl (c) LiCl (d) KCl**

✔ ANSWER

(b) HCl – egg-shells are calcium carbonate; an acid reacts with it to give CO_2, which turns lime water milky.

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

3 **10 mL of NaOH solution is completely neutralised by 8 mL of a given HCl solution. If we take 20 mL of the same NaOH solution, the HCl solution required will be (a) 4 mL (b) 8 mL (c) 12 mL (d) 16 mL**

✔ ANSWER

(d) 16 mL – double the volume of NaOH needs double the HCl: 8 mL × 2 = 16 mL.

4 **Which one of the following types of medicines is used for treating indigestion? (a) Antibiotic (b) Analgesic (c) Antacid (d) Antiseptic**

✔ ANSWER

(c) Antacid – it is a mild base that neutralises the excess acid in the stomach.

5 (a) **Write word and balanced equations: dilute sulphuric acid reacts with zinc granules.**

✔ ANSWER

Sulphuric acid + Zinc → Zinc sulphate + Hydrogen

$H_2SO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + H_2(g)$

5 (b) **Dilute hydrochloric acid reacts with magnesium ribbon.**

✔ ANSWER

Hydrochloric acid + Magnesium → Magnesium chloride + Hydrogen

$2HCl(aq) + Mg(s) \rightarrow MgCl_2(aq) + H_2(g)$

5 (c) **Dilute sulphuric acid reacts with aluminium powder.**

✔ ANSWER

Sulphuric acid + Aluminium → Aluminium sulphate + Hydrogen

$3H_2SO_4(aq) + 2Al(s) \rightarrow Al_2(SO_4)_3(aq) + 3H_2(g)$

5 (d) **Dilute hydrochloric acid reacts with iron filings.**

✔ ANSWER

Hydrochloric acid + Iron → Iron(II) chloride (ferrous chloride) + Hydrogen

$2HCl(aq) + Fe(s) \rightarrow FeCl_2(aq) + H_2(g)$

6 **Compounds such as alcohols and glucose also contain hydrogen but are not categorised as acids. Describe an Activity to prove it.**

✔ ANSWER

- Fix two nails on a cork and place it in a 100 mL beaker.
- Connect the nails to a 6 V battery through a bulb and a switch.

- Pour dilute HCl into the beaker and switch on: the **bulb glows**.
- Repeat separately with glucose solution and alcohol solution: the **bulb does not glow**.

Conclusion: HCl gives H^+ ions in water, which conduct current. Glucose and alcohol do not give H^+ ions, so even though they contain hydrogen they are **not acids**.

7 **Why does distilled water not conduct electricity, whereas rain water does?**

✔ ANSWER

Distilled water is pure and has almost **no ions**, so it does not conduct electricity.

Rain water dissolves gases like CO_2 (and SO_2 etc.) from the air, forming acids that give **ions** (H^+ etc.). These ions conduct electricity.

8 **Why does dry HCl gas not show acidic behaviour in the absence of water?**

✔ ANSWER

Acidic behaviour is due to H^+ ions. HCl gives H^+ (as H_3O^+) ions **only in the presence of water**: $HCl + H_2O \rightarrow H_3O^+ + Cl^-$.

Dry HCl gas does not ionise, so it gives no H^+ ions and shows no acidic behaviour.

9 **Solutions A, B, C, D and E showed pH 4, 1, 11, 7 and 9 respectively with universal indicator. Which solution is**

(a) neutral? (b) strongly alkaline? (c) strongly acidic? (d) weakly acidic? (e) weakly alkaline?

Arrange the pH in increasing order of hydrogen-ion concentration.

✔ ANSWER

(a) Neutral: **D** (pH 7)

(b) Strongly alkaline: **C** (pH 11)

(c) Strongly acidic: **B** (pH 1)

(d) Weakly acidic: **A** (pH 4)

(e) Weakly alkaline: **E** (pH 9)

Lower pH = more H^+ ions. Increasing order of H^+ concentration:

C (11) < E (9) < D (7) < A (4) < B (1)

10 **Equal lengths of magnesium ribbon are taken in test tubes A and B. HCl is added to A and acetic acid (CH_3COOH) to B, in the same amount and concentration. In which test tube will the fizzing occur more vigorously and why?**

✔ ANSWER

Fizzing is more vigorous in **test tube A (HCl)**.

HCl is a **strong acid** and gives more H^+ ions than acetic acid (a weak acid) of the same concentration. So it reacts faster with magnesium and gives hydrogen gas more quickly.

11 **Fresh milk has a pH of 6. How do you think the pH will change as it turns into curd? Explain your answer.**

✔ ANSWER

The pH will **decrease** (go below 6).

When milk turns into curd, **lactic acid** is formed, which makes it more acidic.

12 (a) **A milkman adds a very small amount of baking soda to fresh milk. Why does he shift the pH of the fresh milk from 6 to slightly alkaline?**

✔ ANSWER

Baking soda is basic. A slightly alkaline milk **does not turn sour quickly**, because the acid formed is neutralised. So the milk stays fresh for a longer time.

12 (b) **Why does this milk take a long time to set as curd?**

✔ ANSWER

Curd setting needs the milk to become acidic (lactic acid). The lactic acid formed first gets **neutralised by the baking soda**, so it takes longer for the milk to become acidic enough to set as curd.

13 **Plaster of Paris should be stored in a moisture-proof container. Explain why?**

✔ ANSWER

Plaster of Paris absorbs water (moisture) and changes into **gypsum**, a hard solid mass:

$CaSO_4.½H_2O + 1½H_2O \rightarrow CaSO_4.2H_2O$

Then it is no longer useful, so it is kept in a moisture-proof container.

14 **What is a neutralisation reaction? Give two examples.**

✔ ANSWER

The reaction between an acid and a base to give a **salt and water** is called a neutralisation reaction.

Examples:

- $NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$
- $2KOH(aq) + H_2SO_4(aq) \rightarrow K_2SO_4(aq) + 2H_2O(l)$

15 **Give two important uses of washing soda and baking soda.**

✔ ANSWER

Washing soda ($Na_2CO_3.10H_2O$):

- Used in glass, soap and paper industries.
- Used for removing permanent hardness of water (also as a cleaning agent at home).

Baking soda ($NaHCO_3$):

- Used in baking powder to make bread and cake soft and spongy.
- Used as an ingredient in antacids (also in soda-acid fire extinguishers).

🧪 Group Activity

(I) **Prepare a beetroot indicator. Add 2–3 drops to lemon juice (A), soda-water (B), vinegar (C) and baking soda solution (D). Note the colour change in a table.**

💡 SAMPLE ANSWER

Test tube	Substance	Nature	Colour with beetroot extract
A	Lemon juice	Acidic	Stays red / pinkish-red
B	Soda-water	Acidic (weak)	Stays red / pinkish-red
C	Vinegar	Acidic	Stays red / pinkish-red

D	Baking soda solution	Basic	Changes to purple / yellowish

So beetroot extract works as a natural acid-base indicator.

(II) In a soda-acid fire extinguisher, dilute H_2SO_4 mixes with $NaHCO_3$ solution and a discharge comes out of the nozzle. Direct it on a burning candle. What happens?

✔ ANSWER

The candle **goes out**.

The acid reacts with sodium hydrogencarbonate to give **carbon dioxide**, which does not support burning and cuts off air from the flame.

$2NaHCO_3(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l) + 2CO_2(g)$

🎉 **Chapter 2 done! Revise the answers once more tomorrow.**